

12.471

EE25BTECH11042 - Nipun Dasari

Question:

The solution to system of equations

$$\begin{pmatrix} 2 & 5 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -30 \end{pmatrix} \quad (0.1)$$

is

- 1) 6,2
- 2) -6,2
- 3) -6,-2
- 4) 6,-2

Solution:

We can write given system in the form of an augmented matrix and solve using row reduction.

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ -4 & 3 & -30 \end{array} \right) \quad (4.1)$$

$$R_2 \rightarrow R_2 + 2R_1:$$

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 13 & -26 \end{array} \right) \quad (4.2)$$

$$R_2 \rightarrow \frac{1}{13}R_2:$$

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 1 & -2 \end{array} \right) \quad (4.3)$$

$$R_1 \rightarrow R_1 - 5R_2:$$

$$\left(\begin{array}{cc|c} 2 & 0 & 12 \\ 0 & 1 & -2 \end{array} \right) \quad (4.4)$$

$$R_1 \rightarrow \frac{1}{2}R_1:$$

$$\left(\begin{array}{cc|c} 1 & 0 & 6 \\ 0 & 1 & -2 \end{array} \right) \quad (4.5)$$

From the reduced row echelon form, we get the solution:

$$x = 6 \quad (4.6)$$

$$y = -2 \quad (4.7)$$

The solution vector is

$$\mathbf{x} = \begin{pmatrix} 6 \\ -2 \end{pmatrix} \quad (4.8)$$

